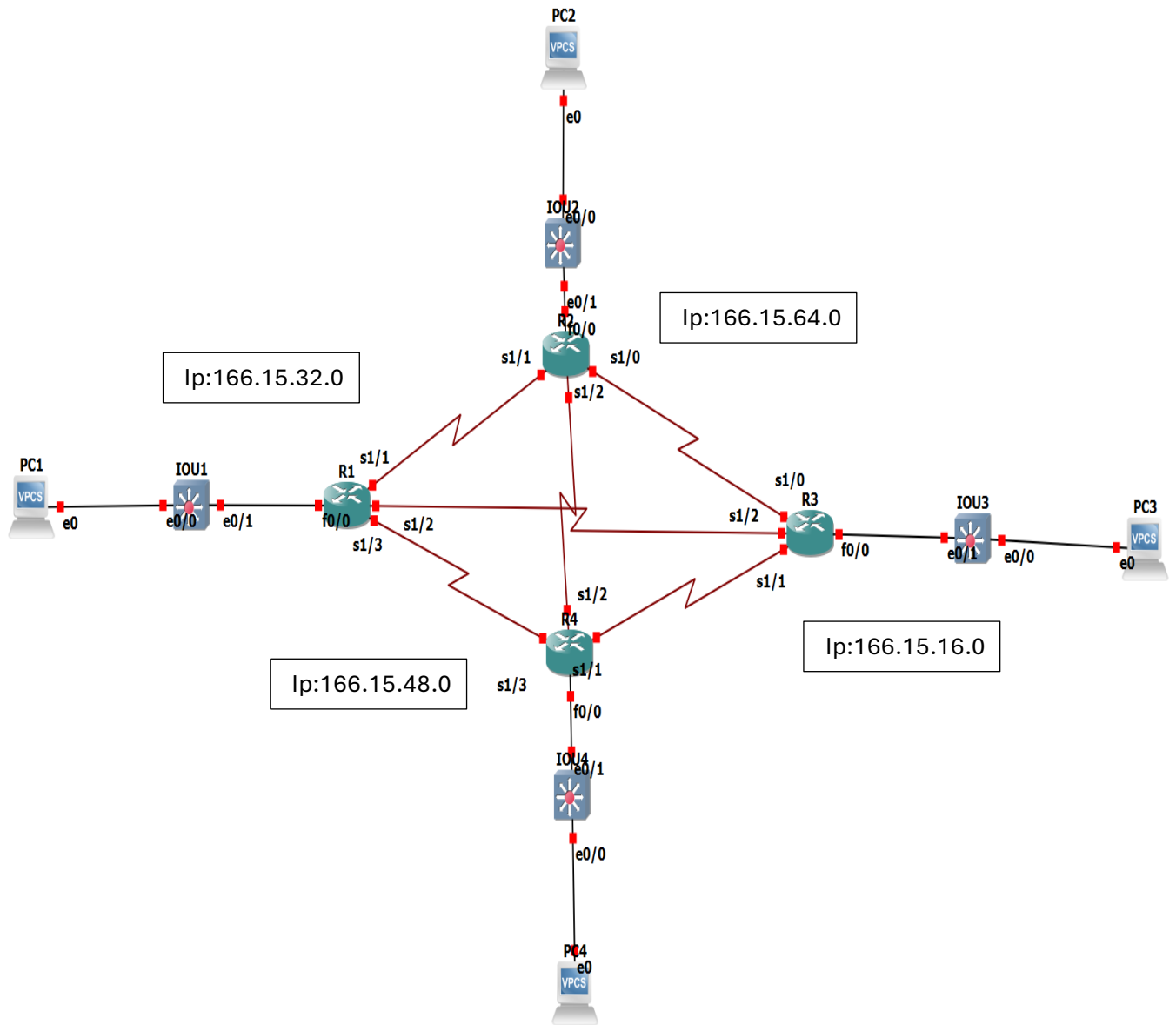




Report

OSPF

Name /Mohammed Talal Alobaysi



Commands

```
R1#conf t
R1(config)#int s1/0
R1(config-if)#ip address 166.15.80.1
R1(config-if)#ip address 166.15.80.1/20
R1(config-if)#ip address 166.15.80.1 255.255.254.0
R1(config-if)#no shut
R1(config-if)#
R1(config)#int s1/1
R1(config-if)#ip address 166.15.112.1 255.255.254.0
R1(config-if)#no shut
R1(config-if)#
R1(config)#ip route 166.15.32.0 255.255.254.0
R1(config)#ip route 166.15.64.0 255.255.254.0 s1/0
R1(config)#ip route 166.15.16.0 255.255.254.0 s1/1
R1(config)#ip route 166.15.48.0 255.255.254.0 s1/1
R1(config)#copy running-config startup-config
R1(config)#exit
```

```
R2#conf t
R2(config)#int s1/0
R2(config-if)#ip address 166.15.80.2 255.255.254.0
R2(config-if)#no shut
R2(config-if)#int s1/2
R2(config-if)#ip address 166.15.128.1 255.255.254.0
R2(config-if)#no shut
R2(config-if)#exit
R2(config)#ip route 166.15.32.0 255.255.254.0 s1/0
R2(config)#exit
```

```
R3#conf t
R3(config)#int s1/1
R3(config-if)# ip address 166.15.112.2 255.255.254.0
R3(config-if)#no shut
R3(config-if)#int s1/3
R3(config-if)# ip address 166.15.160.2 255.255.254.0
R3(config-if)#no shut
R3(config-if)#exit
R3(config)#ip route 166.15.32.0 255.255.254.0
R3(config)#ip route 166.15.32.0 255.255.254.0 s1/1
R3(config)#ip route 166.15.16.0 255.255.254.0 s1/3
R3(config)#exit
```

```
R4#conf t
R4(config)#int s1/2
R4(config-if)#ip address 166.15.128.2 255.255.254.0
R4(config-if)#ip address 166.15.128.2 255.255.254.0
R4(config-if)#no shut
R4(config-if)#int s1/3
R4(config-if)#ip address 166.15.160.1 255.255.254.0
R4(config-if)#no shut
R4(config-if)#exit
R4(config)#ip route 166.15.64.0 255.255.254.0 s1/2
R4(config)#ip route 166.15.48.0 255.255.254.0 s1/3
R4(config)#exit
```

```
R1#conf t
R1(config)#interface loopback 1
state to up
R1(config-if)#ip address 1.1.1.1 255.255.255.255
R1(config-if)#end
R1#conf t
R1(config)#router ospf 10
R1(config-router)#network 166.15.16.0 255.255.254.0 area 0
R1(config-router)#network 166.15.32.0 255.255.254.0 area 0
R1(config-router)#network 166.15.48.0 255.255.254.0 area 0
R1(config-router)#network 166.15.64.0 255.255.254.0 area 0
R1(config-router)#int s1/0
R1(config-if)# ip ospf 10 area 0
R1(config-if)#int s1/1
R1(config-if)# ip ospf 10 area 0
R1(config-if)#exit
R1(config)#router ospf 10
R1(config-router)#passive-interface loopback 1
R1(config-router)#end
R1#show ip protocols
*** IP Routing is NSF aware ***
```

Routing Protocol is "ospf 10"

Outgoing update filter list for all interfaces is not set

Incoming update filter list for all interfaces is not set

Router ID 1.1.1.1

Number of areas in this router is 1. 1 normal 0 stub 0 nssa

Maximum path: 4

Routing for Networks:

166.15.16.0 0.0.1.255 area 0

166.15.32.0 0.0.1.255 area 0

166.15.48.0 0.0.1.255 area 0

166.15.64.0 0.0.1.255 area 0

Routing on Interfaces Configured Explicitly (Area 0):

Serial1/1

Serial1/0

Passive Interface(s):

Loopback1

Routing Information Sources:

Gateway	Distance	Last Update
---------	----------	-------------

Distance: (default is 110)

R1#conf t

R1(config)#int s1/0

R1(config-if)#ip ospf network point-to-point

R1(config-if)#exit

R1(config)#end

R1#conf t

R1(config)#int loopback 1

R1(config-if)#ip ospf network point-to-point

R1(config-if)#END

R1-R2

10	23.422402	166.15.80.1	224.0.0.5	OSPF	80 Hello Packet
11	30.007443	N/A	N/A	SLARP	24 Line keepalive, outgoing sequence 216, returned sequence 215
12	31.711393	N/A	N/A	SLARP	24 Line keepalive, outgoing sequence 216, returned sequence 216
13	32.883048	166.15.80.1	224.0.0.5	OSPF	80 Hello Packet
14	40.005131	N/A	N/A	SLARP	24 Line keepalive, outgoing sequence 217, returned sequence 216
15	41.704137	N/A	N/A	SLARP	24 Line keepalive, outgoing sequence 217, returned sequence 217
16	42.572216	166.15.80.1	224.0.0.5	OSPF	80 Hello Packet
17	48.308066	N/A	N/A	CDP	346 Device ID: R2 Port ID: Serial1/0
18	50.005385	N/A	N/A	SLARP	24 Line keepalive, outgoing sequence 218, returned sequence 217
19	51.706395	N/A	N/A	SLARP	24 Line keepalive, outgoing sequence 218, returned sequence 218
20	51.888015	166.15.80.1	224.0.0.5	OSPF	80 Hello Packet
21	59.999977	N/A	N/A	SLARP	24 Line keepalive, outgoing sequence 219, returned sequence 218
22	61.652106	166.15.80.1	224.0.0.5	OSPF	80 Hello Packet

R1-R4

8	21.688547	N/A	N/A	SLARP	24 Line keepalive, outgoing sequence 219, returned sequence 219
9	28.351935	166.15.112.1	224.0.0.5	OSPF	80 Hello Packet
10	30.000693	N/A	N/A	SLARP	24 Line keepalive, outgoing sequence 220, returned sequence 219
11	31.690822	N/A	N/A	SLARP	24 Line keepalive, outgoing sequence 220, returned sequence 220
12	32.317945	N/A	N/A	CDP	346 Device ID: R3 Port ID: Serial1/1
13	37.503998	166.15.112.1	224.0.0.5	OSPF	80 Hello Packet
14	40.000955	N/A	N/A	SLARP	24 Line keepalive, outgoing sequence 221, returned sequence 220
15	41.691618	N/A	N/A	SLARP	24 Line keepalive, outgoing sequence 221, returned sequence 221
16	44.084989	N/A	N/A	CDP	337 Device ID: R1 Port ID: Serial1/1
17	46.552103	166.15.112.1	224.0.0.5	OSPF	80 Hello Packet
18	49.994166	N/A	N/A	SLARP	24 Line keepalive, outgoing sequence 222, returned sequence 221
19	51.687020	N/A	N/A	SLARP	24 Line keepalive, outgoing sequence 222, returned sequence 222
20	56.013060	166.15.112.1	224.0.0.5	OSPF	80 Hello Packet

Routing Table R1

```

O      166.15.0.0/16 is variably subnetted, 14 subnets, 2 masks
C      166.15.16.0/20 [110/3] via 166.15.96.2, 02:48:52, Serial1/3
C      166.15.32.0/20 is directly connected, FastEthernet0/0
L      166.15.32.1/32 is directly connected, FastEthernet0/0
O      166.15.48.0/20 [110/3] via 166.15.112.2, 02:40:51, Serial4/1
O      166.15.64.0/20 [110/2] via 166.15.80.2, 03:18:37, Serial1/2
C      166.15.80.0/20 is directly connected, Serial1/2
L      166.15.80.1/32 is directly connected, Serial1/2
C      166.15.96.0/20 is directly connected, Serial1/3
L      166.15.96.1/32 is directly connected, Serial1/3
C      166.15.112.0/20 is directly connected, Serial4/1
L      166.15.112.1/32 is directly connected, Serial4/1
O      166.15.128.0/20 [110/6] via 166.15.80.2, 02:53:05, Serial1/2
O      166.15.144.0/20 [110/8] via 166.15.80.2, 02:42:25, Serial1/2
O      166.15.160.0/20 [110/5] via 166.15.112.2, 02:40:02, Serial4/1
O      166.15.160.0/20 [110/5] via 166.15.96.2, 02:42:05, Serial1/3

```

Destination	Next	Cost
N2	R2	2
N3	R3	3
N4	R4	3

Ospf general header:

Version 2	Type 4	Message Length 44
Source router IP address 166.15.32.1		
Area identification		
Checksum		Authentication type
Authentication		
Number of Advertisements 1		

LSA general header:

Link state age 1	DN	O	DC	L	N	MC	E	MT	Link state type 1
Link state ID 166.15.64.1									
Advertising router 166.15.64.1									
Link state sequence number									
Link state checksum						Length			
H	N	W	V			E	B	N of links 3	

LSA header 1:

Link ID 166.15..80.1		
Link data S1/1		
Link type 1	N of Metrics	0 Metric 2

LSA header 2 :

Link ID 166.15.112.1		
Link data S1/3		
Link type 1	N of Metrics	0 Metric 3